Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series: A Signal-Loss Audit on the Pacific Northwest Network

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Abstract

Network filters remove the common-mode error of GNSS position time series by predicting each station from other stations and subtracting the prediction. Every such filter also removes whatever real deformation the other stations share with the target, and the published evaluations of these filters, which rest on the reduction of residual scatter, cannot see this loss. This project proposes a signal-loss audit of network filters on the daily solutions of the continuously operating GNSS stations of the Pacific Northwest of North America (Oregon, Washington, southern British Columbia and northern Idaho), as published by the Nevada Geodetic Laboratory: 540 stations with a median spacing of 6 km above the Cascadia subduction zone, where slow-slip events of a few millimetres recur every 14 months and are the deformation signal that network filtering is most likely to damage. Transients of known extent, amplitude and duration are planted in the real residuals of the network; each filter is run with and without them; and the paired difference gives the fraction of each transient that survives, together with the noise the filter removes, the spurious signal it creates elsewhere and the fraction of each station's seasonal cycle that it keeps. Classical target-inclusive filters, distance-excluded stacks, leave-one-out filters over nearest neighbours and learned filters, including attention networks, are audited under one protocol in which the exclusion radius around the target and the number of stations read are experimental variables. The method was developed on 1,251 stations in California; the Pacific Northwest data set is the study network, with its test period untouched until the method is frozen. The audit will state,

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with ground truth, what each filter costs in signal on a dense network and which settings keep transients of the size of Cascadia slow slip.

Keywords: GNSS time series, common-mode error, spatial filtering, transient deformation, signal preservation, deep learning, Cascadia

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Status of this document. This is a draft research proposal, written on 5 November 2026 for the author to read and rework into a project application. Its text was drafted by Claude Fable 5.1 (Anthropic) through Claude Code, at the author's request, from the repository, its design documents, the internal working report on the California development work and the coverage analyses of the Turkish and Pacific Northwest data. The research questions are the author's. Statements about the Pacific Northwest and Turkish data were measured on series downloaded on 5 November 2026; statements about the California development work describe results obtained with a processing in which the seasonal cycle was removed, which the author has since reversed, so those results are indicative only and are cited without numbers. Nothing in this document has been computed on the study network's test period.

1. Summary

Continuously operating GNSS stations record crustal deformation together with noise, part of which is common to many stations of a network. Network filters estimate this common-mode error from the other stations and subtract it, and are a standard step before the detection of transient deformation, the estimation of velocities and the study of postseismic processes. The literature evaluates these filters almost exclusively by the reduction in the scatter of the filtered series. Scatter reduction, however, rewards the removal of anything the stations share, deformation included, and no study has measured with ground truth how much real signal a filter removes as a function of the signal's size relative to the filter's reach.

The project will carry out such a measurement on the Pacific Northwest network. Its aims are (i) to quantify, for each filter family, the fraction of a transient that survives filtering as a function of the transient's width, amplitude and duration and of the filter's exclusion radius and number of stations; (ii) to measure, on the same runs, the noise removed, the spurious signal created elsewhere, the fraction of each station's seasonal cycle kept, and the retention of real earthquake offsets and of the catalogued slow-slip episodes; (iii) to compare classical, distance-excluded, leave-one-out and learned filters

at equal signal kept rather than on noise alone; and (iv) to deliver, as open software, an audit that any future filter for a GNSS network can be run through.

The study network is the Pacific Northwest of North America: the 540 stations between 42.5° and 49.5°N and 125.5° and 116.0°W with at least three years of daily solutions in the Nevada Geodetic Laboratory archive, covering Oregon, Washington, southern British Columbia and northern Idaho above the Cascadia subduction zone. The network is dense (median nearest neighbour 6.2 km), continuous from 2008 to the present, and records episodic tremor and slip events of 2–5 mm every 14 months, the kind of regional transient that a network filter can absorb. The method, the software and the protocol exist and were developed on 1,251 stations in California; the Pacific Northwest is the network on which they will be applied under a protocol in which the test period is not examined until the method is frozen. The Turkish national network was examined first and set aside, because its daily solutions are publicly available only for 2009–2015 and two short later windows (Section 9).

2. Background and significance

2.1. Common-mode error and its removal

Daily GNSS position time series contain, besides the deformation of the ground, noise that is correlated across a network: errors in satellite orbits and clocks, in the realisation of the reference frame and in the modelling of large-scale environmental loading (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). This common-mode error has been estimated by stacking the residuals of the other stations (Wdowinski et al., 1997), by principal or independent component analysis (Dong et al., 2006; Gruszczynski et al., 2018; Zhou et al., 2022), and by weighted means and medians of neighbouring stations (Tian and Shen, 2016; Kreemer and Blewitt, 2021). Network filtering began with the Southern California Integrated GPS Network, where the common mode was the weighted mean of the residuals of the other stations (Wdowinski et al., 1997); principal component analysis and the Karhunen–Loève expansion extended this to common modes that vary in space (Dong et al., 2006); and later work made principal component filtering applicable to networks whose stations start and stop at different times (Shen et al., 2014; Gruszczynski et al., 2018; Li et al., 2020). Because the common mode varies in space, large networks have been divided into sub-regions (Liu

et al., 2026). Tian and Shen (2016) weighted each neighbouring station by its distance and by the correlation of its residuals with the target and chose the weights by minimising the residual scatter after filtering; Kreemer and Blewitt (2021) take a correlation-weighted median over nearby stations, the target included when it qualifies. Across this family, performance is reported as scatter reduction, reduced inter-station correlation, velocity uncertainty or the recovery of a simulated common mode. Kreemer and Blewitt (2021) themselves note that the main reason their filter reduces scatter more than traditional stacking is that it absorbs local common-mode transients.

2.2. Learned filters

Deep learning has entered the field from two sides. The graph neural network of Bachelot et al. (2025) predicts each station from its eight nearest stations beyond 400 km; the distance was chosen empirically, in Cascadia, as the shortest at which 280 observed slow-slip offsets were no longer reduced by denoising, and a plain stack of the same distant stations performed comparably. Costantino et al. (2024) train graph recurrent layers and transformers to extract slow slip from synthetic examples added to noise synthesised from real series, and later used the denoised data to invert fifteen years of slow slip (Costantino et al., 2026). Mastella et al. (2025) model high-frequency scatter at single stations with synthetic training noise. Teutschmann et al. (2026) pretrained a self-supervised model on 17,651 stations of the Nevada Geodetic Laboratory. Predicting a withheld sensor from its neighbours is an established principle elsewhere, in ocean-bottom seismology (Crawford and Webb, 2000; Janiszewski et al., 2019), distributed acoustic sensing (Batson and Royer, 2019; van den Ende et al., 2023) and spatiotemporal kriging (Wu et al., 2021; Cini et al., 2022; Marisca et al., 2022), and excluding nearby observations so that spatial autocorrelation does not inflate apparent skill is the basis of buffered spatial cross-validation (Le Rest et al., 2014).

2.3. How signal preservation has been assessed

Three approaches recur. Observed transients can be compared before and after filtering (Bachelot et al., 2025); this uses real signals but has no ground truth. Synthetic signals can be added to synthetic noise (Costantino et al., 2024; Mastella et al., 2025); this has ground truth but depends on how realistic the noise is, and has been analysed against signal-to-noise ratio rather than spatial scale. Independent data, such as tectonic tremor, can confirm that an extracted signal is real (Costantino et al., 2024), but cannot

measure what was lost. In Cascadia, Costantino et al. (2026) report that the scaling, b-value and stress-drop distribution of slow-slip events change with the detection threshold applied to slip inverted from denoised data, and the debate over a reported precursory acceleration of slip before large earthquakes (Bletery and Nocquet, 2023, 2025) turned on whether a common-mode correction removes only noise. The question of what a filter removes is therefore not a technicality: it decides which published deformation signals are real.

2.4. Gaps

Four gaps motivate the project. No study has measured, with ground truth, how much of a transient a common-mode filter removes as a function of the transient's size relative to the filter's reach. The reach has been chosen empirically, as a fixed 400 km (Bachelot et al., 2025) or by scatter-based thresholds (Tian and Shen, 2016). Filters that use the target station, which include most classical methods, have not been compared with leave-one-out filters on signal loss. And learned filters have been compared with simple ones on noise reduction, not at equal signal kept. A fifth gap is specific to this project: the filters developed for Cascadia (Bachelot et al., 2025; Costantino et al., 2024) and the catalogues built from the denoised data (Costantino et al., 2026) have been evaluated on observed slow-slip offsets without ground truth, and no measurement exists of how much of a transient of the size of a Cascadia slow-slip event these filters, or the classical ones, remove.

2.5. Significance of the study network

The Cascadia subduction zone produces the deformation signals that network filters are most likely to damage: episodic tremor and slip events that move the surface by 2–5 mm over two to four weeks along hundreds of kilometres of the margin (Michel et al., 2019), and the long-term afterslip and viscoelastic relaxation of the plate interface. Every GNSS study of these processes filters the series first, and the recent debate over precursory slow slip (Bletery and Nocquet, 2023, 2025) and over the statistics of slow-slip catalogues derived from denoised data (Costantino et al., 2026) turns on what the filtering removed. A dense, continuous and openly processed network over such a source is the setting in which a signal-loss audit is both feasible and consequential: the filters developed and used in this region (Bachelot et al., 2025; Costantino et al., 2024) can be audited on the network they were built

for, and the recommended settings apply directly to the catalogues that are built from it. The audit also establishes, for a sparse national network such as Türkiye's, what would have to be measured once its data become available (Section 9).

3. Aims and research questions

The project tests one hypothesis and answers four questions.

Hypothesis. The fraction of a transient that a network filter removes is set by the share of the filter's input that lies inside the transient: a filter over the nearest stations removes most of any transient wider than their spacing, while a filter over the whole network loses only the share of the transient that enters the network mean, which reappears with opposite sign at every other station.

Questions.

1. For each filter family, how does the retention of a planted transient depend on its width, amplitude and duration, and on the filter's exclusion radius and number of stations, on a dense network above a subduction zone?
2. At equal signal kept, which filter removes most noise, and does any learned filter remove more than the unweighted mean of all distant stations?
3. How much of a station's seasonal cycle, of real coseismic offsets and of real slow-slip episodes does each filter keep, and how far does the spurious signal of a real event reach through each filter?
4. Do the answers obtained on the Californian network, where the method was developed, hold on a second dense network with a different tectonic setting and noise, or are the trade-offs between noise removed and signal kept network-specific?

4. Data

4.1. The Pacific Northwest stations in the Nevada Geodetic Laboratory archive

The data are the daily IGS20 solutions of the Nevada Geodetic Laboratory (NGL) (Blewitt et al., 2018; Nevada Geodetic Laboratory, 2026), which

Table 1: The study network, as measured on the series downloaded on 5 November 2026 and on the residual array built from them.

Stations selected (box, $\geq$ 1,095 solutions)	54
Median nearest-neighbour distance	6.2 k
Stations with $\geq$ 300 solutions in 2008 / 2013 / 2019 / 2020–2024	231 / 318 / 356 / 354–38
Stations kept by the build	47
Days	6,872 (2008-01-01 to 2026-10-2
Station-day coverage	69
Held-out stations (10% by farthest-point sampling, plus short records)	7
Training cells barred by the slow-slip episode exclusion	14
Median stations within 300 km at $R = 0$ / 400 km	244 / 13

processes the public RINEX files of continuously operating stations worldwide with one software and one reference frame. Each solution gives the east, north and up position of a station's antenna relative to a fixed reference point near the station, estimated from 24 hours of observations, with formal uncertainties of about 1 mm in the horizontal and 3–4 mm in the vertical. Using the NGL archive rather than an in-house processing has two advantages for an audit: every station is processed identically, and the filters audited here are the filters that users of this public archive apply.

All NGL stations between 42.5° and 49.5°N and between 125.5° and 116.0°W with at least 1,095 daily solutions were selected on 5 November 2026: 540 stations of the Pacific Northwest Geodetic Array, the Network of the Americas and other networks. Their median distance to the nearest station is 6.2 km (quartiles 2.1 and 15.0 km; California 9.5 km). The series start in 2008 at the median station and run to the present; 231 stations have at least 300 daily solutions in 2008, 318 in 2013, 356 in 2019 and 354–387 in every year from 2020 to 2024. A median of 92% of each station's span has a solution, and 98 stations have a gap longer than a year, which the availability mask records. The residual array built on 5 November 2026 keeps 472 of the 540 stations (5 without a baseline, 16 removed by the station quality control and 47 co-located twins), with 6,872 days from 1 January 2008 to 24 October 2026, a coverage of 69% of the station-days, and a median of 244 stations within the 300-km neighbour cap of each station (Table 1). The training exclusion of the GNSS-catalogued slow-slip episodes (Section 5.4) touches 278 stations and bars 14% of the training cells.

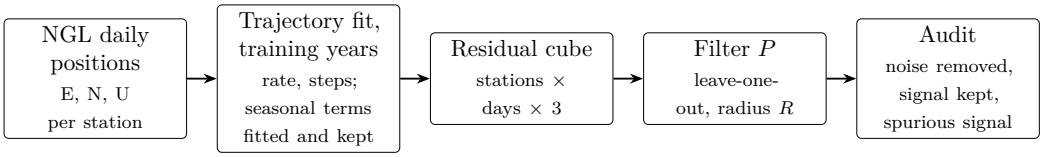


Figure 1: Processing chain. Planted transients are added to the residual cube before filtering, and the paired difference of the cleaned series gives the fraction kept.

4.2. Consequences for the design

The data allow the same design as the Californian development, with the training, validation and test years of Section 5.6 and planted transients from 10 to 100 km in width at exclusion radii from 0 to 400 km. The network differs from California in three ways that make it a test rather than a repetition. Its deformation is dominated by a subduction zone rather than by strike-slip faults, so the real transients are wide, slow and coherent along the margin, which is the regime in which averaging filters absorb signal. Its noise, under a maritime climate, has a different seasonal and loading content. And its station density is highest along the coast and the Puget Lowland and lowest inland, so a station's nearest neighbours span a wider range of distances than in California.

5. Method

The method is implemented in the open `gnssdl` package, written and tested during the Californian development (149 automated tests, including the leak checks of Section 5.2). Figure 1 shows the processing chain: daily positions are reduced to residuals by removing each station's trajectory, screened, and stored in a single array; every filter is applied to this array; and the cleaned series are scored by the audit.

5.1. Residuals

For each station i and component c, the trajectory model

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[p_k \sin(2\pi k t) + q_k \cos(2\pi k t)\right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t) \tag{1}$$

is fitted by least squares on the training years (for a station with fewer than 365 daily solutions in them, on the first two years of its record, or on its

first 365 solutions if these reach further; such stations are always among the held-out ones), where a is a constant, v a linear rate about the mean epoch t_0 of the fitted data, p_k and q_k the amplitudes of the annual and semi-annual harmonics, b_j the amplitudes of Heaviside steps H at the epochs t_j of the NGL step catalogue and t the time in years. The fitted trajectory is extrapolated to the later years. The residual $r_{i,c}(t)$ is the series minus the constant, the rate and the equipment steps. The seasonal terms are estimated so that an incomplete cycle does not bias the rate, but they are added back: the residual carries each station's seasonal cycle, and what a filter does to it is measured directly (Section 5.3). Earthquake steps are likewise estimated and added back, so that coseismic offsets, which are spatially coherent deformation of the kind a filter should preserve, remain in the series. Equipment steps are removed everywhere, within the training years in the fit and after them by a local estimate, the median residual over the 30 days after the change minus that over the 30 days before. A value more than five robust standard deviations from its 61-day running median is treated as missing. For station screening only, a second fit is made from which the earthquake steps and the seasonal terms are also removed, so that real offsets and the seasonal cycle do not count as noise. A station is dropped if the robust scale of this fit exceeds 10 mm in the horizontal or 30 mm in the vertical, which removes non-tectonic site motion, and stations within 1 km of a station with more daily solutions are dropped as duplicates. The second fit changes no series: the residual passed to the filters keeps the seasonal cycle. The residuals, their uncertainties and an availability mask are stored with the neighbour lists of every station at each exclusion radius.

5.2. Leave-one-out filtering and the exclusion radius

A filter P estimates the part of the residual of station i that is shared with other stations, and the cleaned series is $r - P(r)$. A leave-one-out filter never reads station i itself when estimating it, and never reads any station within R km of it. The exclusion radius R and the number of stations read are experimental variables, the far stack runs at $R = 0$, 25, 50, 100, 200, 400 and 600 km, the nearest-neighbour and learned filters at the cube's radii of 0, 10, 25, 50, 100, 200 and 400 km (the whole-network M6 variants up to 100 km), and the radii reported for the test years are fixed on the validation years before the test period is touched. Target-inclusive filters, which read the target on purpose as published filters of their kind do, are applied to the

same data and scored on noise removed and on the signal measures, but not on reconstruction error.

Two leak checks are run for every leave-one-out filter at every radius. In the radius check, the target and every station within R of it are set to a poison value of 10^6 mm, for five randomly chosen targets; in the mask check, cells hidden from a filter are filled once with missing values and once with the poison value. In both, the predictions must agree to within 0.001 mm. A filter that passes never reads what it must not, and one prediction on the unhidden data is then a leave-one-out prediction for every station at once.

5.3. Measures

Filters are compared on two measures of noise and five of signal, all computed from the same runs; the planted transients are the primary signal test, and the real signals are secondary checks without ground truth.

Noise removed is the reduction in short-period scatter at each scored station, $1 - \mathrm{rms}(\mathrm{fast}(r - P(r)))/\mathrm{rms}(\mathrm{fast}(r))$, where the fast part of a series is the series minus its 61-day running median; it is averaged over the three components and summarised by its median over stations. Predicting zero gives 0. The measure covers periods shorter than about two months.

Reconstruction error is the root mean square of the fast part of the leave-one-out error $r - P(r)$, in units of each station's robust noise scale. It is the criterion on which the leave-one-out filters are tuned, on the validation years. Because the error is lower when the prediction reproduces whatever the target shares with other stations, signal included, tuning on it favours settings that absorb coherent signal; the signal measures are therefore independent checks on what the tuning selected, not refinements of it.

Retention measures the fraction of a planted transient that survives filtering. A transient s is added to the residuals, both the original and the modified residuals are filtered, and the change in the cleaned series and the retention are

$$\Delta = \Big(r + s - P(r+s)\Big) - \Big(r - P(r)\Big), \qquad \rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle}, \tag{2}$$

with the inner products over the horizontal components, the days of the transient and the scored stations within twice its width of its centre. Pairing with the unmodified run removes the noise term; for a linear filter, $\rho = \langle s - P(s), s \rangle / \langle s, s \rangle$ exactly. $\rho = 1$ means that the transient is kept, $\rho = 0$ that it is removed, and $\rho > 1$ that the filter adds to it.

Spurious signal is the change Δ created at scored stations outside the transient's footprint, reported as the largest peak horizontal amplitude in units of the planted amplitude, together with the number of stations affected and the sum of their peak changes, since a network-wide filter spreads a small ghost over every station while a local filter creates a larger one at a few.

Seasonal kept fits, per scored station and component over the period, constant, rate, annual and semi-annual terms and Heaviside steps at the station's catalogued earthquakes to r and to $r - P(r)$ on the same days, and reports $\langle c_{\mathrm{kept}}, c_{\mathrm{obs}}\rangle / \langle c_{\mathrm{obs}}, c_{\mathrm{obs}}\rangle$ over the four harmonic coefficients c. A station's seasonal cycle is nearly the same across a network, so it is the limiting case of a wide transient; a filter that averages the network is expected to remove most of it, and the measure states how much of each station's own cycle survives.

Real offsets are every catalogued earthquake step in the scored period at a scored station with an observed horizontal offset of at least 3 mm and no other step within 15 days; the offset is the mean position over the 10 days after the earthquake minus that over the 10 days before, and the fraction kept is the same projection computed on the cleaned series.

Slow-slip episodes are scored from a fixed list (Section 5.4): for each episode, the horizontal displacement at the scored stations within its latitude range and within 150 km of the tremor band, as the mean position over the 10 days after its end minus the mean over the 10 days before its start, is projected on the same quantity in the cleaned series; the median over stations and episodes is reported, with the tremor-timed episodes as a separate group. The episodes of the training years are excluded from training.

5.4. Planted and real signals

Gaussian transients have the form $s(d,t) = A \exp(-d^2/2L^2)\, g(t)\, \mathbf{u}$, where d is the distance of a station from the centre, A the amplitude, L the width, $\mathbf{u}$ a horizontal unit vector in a random direction and $g(t)$ a raised-cosine rise over D days, a 60-day hold and a raised-cosine fall, so the transient leaves no permanent offset. The grid is $A \in \{2, 5, 10\}$ mm, $D \in \{10, 30, 90\}$ days and $L \in \{10, 25, 50, 100\}$ km, each combination planted at 20 centres chosen among the scored stations, with many transients sharing one run of a filter where no station can be changed by two of them.

The audit's signal tests are these planted transients. A library of realistic fault-slip events, built in the Californian development from published source

models placed on catalogued faults, is not used in this project; it may be added later for the Anatolian faults, and the software supports it.

Real signals, as secondary checks, are the catalogued earthquake offsets of the scored period and the Cascadia slow-slip episodes. A fixed episode list has been prepared and committed: 24 episodes from 2007 to 2017 from the GNSS catalogue of Michel et al. (2019), with their along-strike extents from the paper's source data, and 31 episodes from 2022 onward clustered from the tremor catalogue of the Pacific Northwest Seismic Network with fixed rules, which are reported as a separate group. The 2007–2017 episodes fall in the training years and are excluded from training, as the Ridgecrest postseismic interval was in California, so that no filter is fitted on them (Section 4). These observed signals have no ground truth and complement the planted ones.

5.5. Filters

Table 2 lists the filters. C1 is the stack of Wdowinski et al. (1997); C3 is sub-regional probabilistic principal component analysis (Tipping and Bishop, 1999; Gruszczynski et al., 2018; Liu et al., 2026) with 1, 4 or 8 regions and 1–3 components; M2-self is the robust weighted median with the target included, as in common-mode component imaging (Tian and Shen, 2016; Kreemer and Blewitt, 2021). The far stack is the unweighted mean of every station beyond R, the target excluded, and is the parameter-free baseline; D1 is the far stack at 400 km, the distance of Bachelot et al. (2025). M1 is the distance-weighted mean of the nearest stations, M2 its robust weighted median after outlier screening (Kreemer and Blewitt, 2021), M3 ridge regression on the neighbours' components per station or pooled, and M4 the same per period band. The learned filters are M3k, a kernel whose neighbour weights are learned from distance, azimuth and noise level but never from the values; M5, a space–time attention network that predicts a correction to the far stack from the target's neighbours and the far stack over a 64-day window; and M6, an attention network over a wider set of stations or the whole network, in which geometry enters only as a learned bias from pairwise distance and azimuth. Each learned filter is also trained with planted transients in its inputs but not in its targets (the "-aug" versions), which in the Californian development was the training that stopped the network from absorbing coherent signal. Learned filters are trained on quiet residuals, the residuals minus the earthquake offsets estimated in the fit, on training days outside the excluded slow-slip episodes and outside ± 30 days of $M \geq 6$ earthquakes, with early stopping on the validation reconstruction error and three training

seeds. The number of neighbours is 16, 64 and 256 for M1 and M6 and 16 and 256 for M3k; M2 chooses its number on the validation years, M3 and M4 read 16 and M5 chooses from 64.

5.6. Protocol

The days are split into training (2008–2019), validation (2020–2021) and test years (2022 onward); days published after the freeze form a prospective block. Ten percent of the stations are held out by farthest-point sampling (Gonzalez, 1985) and never used in training, so that the generalisation of learned filters to unseen stations is measured. All design decisions, tuning and model selection are made on the validation years. The method is then frozen, with a tagged version of the software, and the test years are scored once, with the settings chosen on validation; no result from them is examined before the freeze, and the project's statements about them are made from that single scoring. Differences between filters are tested with a paired block bootstrap over stations and 30-day blocks for reconstruction error, and by resampling the planted transients, paired by identifier, for retention; a difference whose interval includes zero is reported as not distinguishable. A filter is reported as better than the far stack only if its error is lower with an interval excluding zero and its retention of 100-km transients and its spurious signal are within 0.05 of the far stack's; a filter that fails the signal conditions is reported as a noise gain bought with signal, whatever its error.

6. Preliminary work

The method, the software and the protocol were developed between October and November 2026 on the NGL daily solutions of 1,251 stations in and around California, with seventeen filter configurations from the four families above. That development established the audit's measures, the leak checks, the planted-transient tests, the real-offset and postseismic tests and the bootstrap comparisons, and it produced three findings that shape this proposal. First, the retention of a planted transient by the mean of all stations matched, within a few hundredths, the share of stations the transient covers, computed from the station positions alone; and filters over the 16 nearest stations removed most of any transient wider than their spacing. Second, no learned filter reduced the reconstruction error below that of the far stack with an interval excluding zero, and every learned filter trained without planted transients kept less signal than the far stack; training with planted transients in

Table 2: Filter configurations to be audited. Stations read: the other stations a prediction for the target uses; "nearest" means the nearest stations beyond R. Days used: days of other stations read by a prediction for day t.

Name	Description	Stations read	Days used
Target-inclusive			
C1	Unweighted mean of the whole network	All + target	t
C3	Sub-regional probabilistic PCA, several settings	Region + target	t
M2-self	Robust weighted median of the neighbours and the target	Nearest + target	t (31 d)
Distance-excluded			
Far stack	Unweighted mean of every station beyond R	All beyond R	t
D1	Far stack at $R = 400$ km	All beyond 400 km	t
Leave-one-out over nearest neighbours			
M1	Distance-weighted mean	16, 64, 256 nearest	t
M2	Robust weighted median after outlier screening	Nearest	t (31 d)
M3a, M3b	Ridge regression per station, and pooled with neighbour geometry	Nearest	t
M4a, M4b	M3 per period band	Nearest	All
Learned			
M3k	Learned kernel from distance, azimuth and noise level	Nearest	t
M5, M5-aug	Space–time attention, correction to the far stack	Nearest + far stack	64 d
M6, M6-aug	Attention over many stations or the whole network, with and without the far-stack correction	N nearest or all	64 d

the inputs brought the attention networks' retention up to, not beyond, the far stack's. Third, scatter reduction and reconstruction error rewarded filters that absorbed signal or the target's own noise, so a comparison of filters that omits signal kept and spurious signal can rank a filter that removes signal above one that does not. These results were obtained with a processing that removed the seasonal cycle before filtering, which has since been reversed in favour of keeping it and measuring it; they are therefore cited here as the development evidence for the design, without their numbers, and the Californian data set is development data only. The Pacific Northwest series have been downloaded, their coverage measured and the residual array built (Section 4); the reference filters are being run on its validation years, and nothing has been computed on its test years. The Turkish national network was examined as a candidate study area and set aside (Section 9).

7. Work plan

1. *Data and processing (months 1–2).* Review the residual array built with the processing of Section 5.1 and its dropped stations; confirm the training, validation and test years, the held-out stations and the excluded slow-slip episodes; rebuild if the review changes any of them.
2. *Reference and leave-one-out filters (months 2–4).* Run the target-inclusive, distance-excluded and nearest-neighbour filters at every radius on the validation years; tune their settings; run the leak checks.
3. *Learned filters (months 4–8).* Train the learned kernel and the attention networks, with and without planted transients, at every radius and with three seeds; select on validation reconstruction error; examine the spread across seeds.
4. *Signal tests on validation (months 6–9).* Planted transients, seasonal kept, the real offsets and the slow-slip episodes of the validation years for every filter and radius; the trade-off between noise removed and signal kept; comparison with the Californian development results (question 4).
5. *Freeze and held-out scoring (month 10).* Freeze the method and software; score the test years once, including the real offsets and the slow-slip episodes of 2022 onward.
6. *Reporting (months 10–12).* Manuscript, open release of the software, the episode and event lists, the results tables and the trained models.

8. Expected outcomes

The project will deliver: (i) retention, spurious signal and noise removed as functions of transient width and exclusion radius for every filter family on the Pacific Northwest network, with bootstrap intervals; (ii) the fraction of each station's seasonal cycle, of the real earthquake offsets and of the Cascadia slow-slip episodes kept by each filter, the episodes of the test years measured on data never seen during development; (iii) a comparison of learned and classical filters at equal signal kept, and a statement of whether any learned filter beats the mean of all distant stations on this network; (iv) a direct comparison of two dense networks in different tectonic settings, since the same software and measures were applied to both; (v) a recommended filter and exclusion radius for studies of slow slip and postseismic deformation in Cascadia, stated with the signal cost of the alternatives; and (vi) the open `gnssdl` package, through which any future filter or network, the Turkish one included, can be audited with the same protocol.

9. Risks and limitations

The Turkish national network was the author's first choice of study area and was set aside after its coverage was measured on 5 November 2026. Of the 312 NGL stations between 35.5° and 42.5°N and 25.5° and 45.0°E, 110–165 report daily solutions only from 2009 to October 2015, from September to December 2020 and from September 2022 to April 2023; in the other years 10–35 stations report, and the raw observations for those years are not publicly available. The dense window is 6.7 years long, with a median station spacing of 63 km, so that transients narrower than about 50 km cover a single station and cannot be evaluated, and the learned filters would have about twenty times fewer station-days to train on than in California. The audit's software applies to that network unchanged, and the project will state what its results imply for a 60-km network; a full audit of Türkiye waits for its data.

On the Pacific Northwest, the risks are these. The real slow-slip episodes have no ground truth, and the episodes of 2022 onward are timed from tremor rather than from GNSS, which is why they are reported separately. Ten percent of the stations end before mid-2019 and 98 have gaps longer than a year, so the availability mask matters more than in California; the measures are defined on available days only. The planted transients are Gaussian

in space and symmetric in direction, and do not reproduce the two-sided displacement field of fault slip; the realistic event library of the Californian development is left for later work. Noise removed is a fast-band measure, and the long-period common mode is reached only through the seasonal and episode measures. The learned filters' training choices were made on California and are carried over; a filter that fails here may do so for lack of tuning rather than in principle, and the project reports both the carried-over and the retuned settings.

10. Data, code and use of AI tools

The daily solutions are publicly available from the Nevada Geodetic Laboratory (Nevada Geodetic Laboratory, 2026). The `gnssdl` package is openly available at `https://github.com/hogib/gnssdl` under the MIT licence, with its tests, design documents and the figure code; the results tables, trained models, planted-signal catalogues and the frozen version of the software will be released with the manuscript. The software, its tests and this proposal's text were written by Claude (Anthropic) through Claude Code under the author's direction; the research questions, the design decisions and the choice of study area are the author's.

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